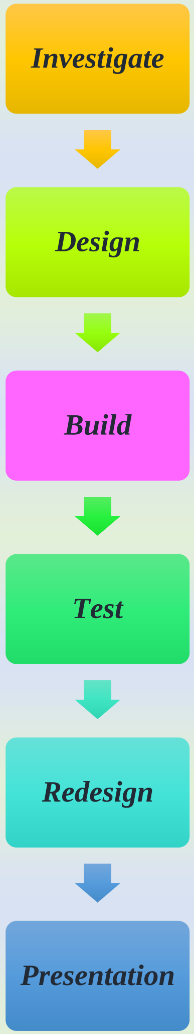
Unit Title: ***Power Smart: Redesigning Our Homes for Energy Efficiency***

Learning Objective	STEM	**Define the Problem (Context)**
Explain the difference between renewable and non-renewable energy and how solar energy can be converted into usable forms. **Identify** common sources of energy loss in homes and propose ways to reduce them. **Describe** how materials affect heat retention and energy efficiency. **Design** a working model of an energy-efficient home that uses solar energy and conserves power effectively. **Solve** mathematical problems related to surface area, insulation material cost, and energy consumption. **Calculate** and graph energy usage and potential savings based on proposed changes.	**Science Content** • Forms and sources of energy • Heat transfer and insulation • Energy transformation and storage • Conductors and insulators • Environmental impacts of energy use **Mathematics Content** • Measurements and unit conversion • Surface area and volume (3D shapes: cube, cuboid, cylinder) • Ratio, proportion, and percentages • Data collection and graphing • Energy cost calculations	**Pakistan is facing an alarming energy crisis.** Power outages are frequent, especially in rural and urban residential areas. The increasing cost of electricity, over-dependence on fossil fuels, and insufficient infrastructure have led to discomfort, economic loss, and reduced access to essential services like internet and cooling/heating systems. To address this, students are challenged to design an **energy-efficient smart home model** that minimizes energy consumption and maximizes the use of **renewable energy sources**, particularly solar power. The goal is to **engineer a smart, sustainable home** for the future using affordable and context-relevant technologies.

STEM Project Planning Checklist

Learning Tools	**Assessment Tools**	**Format of the Unit**	**Resources Needed**	**Venues for the Project**	**References & Support Materials**
(Used to support student exploration, creativity, and construction) 1. **Science Tools**	**Formative Assessment** • Observation checklists during group work	**Project-Based Learning (PBL)** approach • Cross-curricular integration (science, math, ICT, art)	**Physical Materials** • Craft supplies (glue, tape, scissors, markers)	**In-Classroom:** Group collaboration, research, discussion	**National Curriculum Documents** • Pakistan's National Curriculum for

• Thermometers (to test insulation) • Multimeters or voltmeters (to measure electricity) • Solar panels or simple circuits 2. **Math Tools** • Rulers, measuring tapes (for scale modeling) • Graph paper, calculators, protractors • Laptops/tablets for data analysis 3. **Technology Tools** • Internet access for research • Simple coding kits (e.g., Arduino, Micro:bit – optional) • Data loggers or sensors (motion, light, temp) 4. **Engineering/Design Tools** • Cardboard, glue, scissors, box cutters	• Peer and self-assessment rubrics • Exit tickets (daily reflections) • STEM journal entries **Summative Assessment** • Final prototype evaluation rubric (criteria: functionality, creativity, sustainability) • Oral presentation or poster explanation • Data analysis report or reflection paper • Community showcase feedback forms **Rubrics Should Include:** • Understanding of science/math concepts • Design thinking and problem-solving	• Inquiry-based activities and team collaboration • Student-centered: hands-on and minds-on • Structured in phases: 	• Recycled materials (cardboard, plastic, fabric) • Electronics (LEDs, batteries, resistors, breadboards – if available) **Digital Tools** • Videos on energy efficiency and smart homes • Simulators (e.g., Tinkercad for circuits) • Google Sheets or Excel for data analysis • Canva or PowerPoint for presentation design **Learning Supports** • Energy-saving charts and posters • Local electricity bill samples	**STEM Lab or Science Room:** Building and testing prototypes **Outdoor/School Yard:** Solar experiments or airflow tests **Computer Lab:** Research, digital design, simulations **Showcase Venue:** Multipurpose hall, classroom corner, or community room for final presentation	Science and Math (NCP 2022) **Web Resources** • NASA Climate Kids • TeachEngineering.org • Energy.gov – Energy Saver **Books & Guides** • "Engineering is Elementary" (Museum of Science, Boston) • "Design Thinking for Educators" by IDEO • Local textbooks aligned with NCP **Community Resources** • Invite guest speakers: engineers, architects, or energy conservationists • Collaborate with local energy companies or NGOs for context and support

• Recycled materials (bottles, foam, plastic) • LEDs, wires, batteries	• Communication and collaboration • Innovation and creativity • Real-world application and sustainability		• Data usage logs (mock or real)		

<u>Defining the problem:</u>

Lesson 01:

Lesson-Specific Driving Question	**Lesson Summary**	**Learning Outcomes**	**Learning Tasks**	**Assessment Tasks**	**Resources**
If we have enough energy, why should we worry about saving it?	Students are introduced to the growing energy crisis in Pakistan, particularly focusing on residential areas where frequent power outages disrupt daily life. Through videos and articles, they examine the country's reliance on fossil fuels, its limited renewable infrastructure, and the	• Recognize Pakistan's energy crisis and the impacts on everyday life. • Understand the difference between renewable and non-renewable energy. • Identify causes of energy loss in homes. • Begin exploring solutions like solar energy, insulation, and energy-efficient materials.	**Think–Pair–Share:** After viewing a short documentary or reading an article on energy challenges in Pakistan, students discuss: *"Why is electricity often unavailable, and what are the effects?"* **Activity:** Students analyze sample energy bills and identify peak usage patterns.	• Student reflections in journals on the energy crisis and their initial thoughts about solving it. • Responses during group discussion or worksheet showing	• Short video or news clip on Pakistan's energy crisis • Articles: ➢ "Why Does Pakistan Face Power Outages?" ➢ "Energy Wastage in Homes" • Sample energy bills (mock or real) • STEM Challenge Brief (from the Word doc) • Access to research tools (books, internet)

	resulting environmental impact. They are then challenged to investigate how energy-efficient home design and solar power can address this crisis. Students will explore existing smart technologies that reduce energy loss in buildings and evaluate why energy-saving behaviors are essential, even when electricity is available.		**STEM Challenge Introduction:** Present the main design challenge: *“Redesign a model home to be energy-efficient using solar power and sustainable materials.”* **Exploration Task**: In groups, students research common sources of energy loss (e.g., poor insulation, inefficient appliances) and how smart technologies or solar panels help.	understanding of: • Why energy conservation matters Pakistan’s specific energy challenges **<u>Exit ticket:</u>** *“One thing I learned today and one question I still have.”*	

<u>Learn about the problem:</u>

Lesson 02: <u>What is Solar Energy and How Can We Use It in Homes?</u>

Driving Question:	**Lesson Summary:**	**Learning Outcomes:**	**Learning Tasks:**	**Assessment Tasks:**	**Resources:**
How does solar energy power our homes?	Students observe how solar panels convert sunlight into electrical energy. They explore real-world uses of solar energy in households, from lighting to heating water, and investigate	• Understand the process of converting sunlight into usable electricity. • Identify common home appliances	• Watch a short video/demo of how solar panels work. • Analyze case studies of solar-powered homes (local/international).	• Student worksheet: “How Solar Power Works” • Group presentation: “3 Reasons We Should Use Solar Power at Home”	• Video: “Solar Energy for Kids” • Diagram handout of a solar home system • Labeled poster or 3D model of a solar panel

	the benefits and limitations of solar power.	powered by solar energy. • Discuss the advantages of solar energy in the Pakistani context.	• Complete a diagram labelling task of a solar home system.		

Lesson 03: <u>Why Do Homes Lose Energy?</u>

Driving Question:	**Lesson Summary:**	**Learning Outcomes:**	**Learning Tasks:**	**Assessment Tasks:**	**Resources:**
Where and how do homes lose energy?	This lesson helps students identify common points of energy loss (e.g., poor insulation, drafts, inefficient bulbs). They also learn about conductors, insulators, and how materials affect heat transfer.	• Understand how heat escapes through walls, windows, and roofs. • Identify materials that are good insulators or conductors. • Relate energy loss to electricity bills and comfort.	• Experiment: Compare temperature retention in insulated vs. non-insulated boxes. • Conduct a classroom audit: “Where is our classroom losing energy?” • Brainstorm how to reduce energy loss in a typical Pakistani home.	• Observation during experiments • Worksheet: “Insulation and Energy Loss” • **Exit ticket:** “One way to reduce energy waste at home”	• Thermometers, foil, fabric, cardboard • “Energy Loss in Buildings” Worksheet • Infrared image samples (real or printed)

Lesson 04: <u>How Do We Make Homes Energy-Efficient?</u>

Driving Question:	Lesson Summary:	Learning Outcomes:	Learning Tasks:	Assessment Tasks:	Resources:
What makes a home energy-efficient?	Students explore smart home technologies and passive design strategies (e.g., shading, ventilation). They evaluate which solutions are affordable and practical for homes in Pakistan.	• Learn about technologies that conserve power (e.g., motion sensor lights). • Analyze home design strategies for reducing cooling/heating loads. • Evaluate solutions based on cost, sustainability, and context.	• Case study carousel: Smart homes from around the world. • Decision-making activity: “What features should go in a Pakistani smart home?” • Group discussion: “Solar energy + smart design = Sustainable living?”	• Team chart: Must-have vs. nice-to-have energy-saving features • Reflection journal: “What did I learn from today’s smart home designs?”	• Printed profiles of smart homes (with pictures & info) • Chart templates for group work • Local electricity consumption data

Lesson 05: <u>How Can We Measure and Model Energy Use?</u>

Driving Question:	Lesson Summary:	Learning Outcomes:	Learning Tasks:	Assessment Tasks:	Resources:
How do we calculate and visualize energy usage?	This math-integrated lesson guides students to measure, calculate, and graph electricity usage and cost. They explore how proposed energy-saving features affect usage and bills.	• Measure power consumption and convert units • Calculate energy use and cost • Represent findings with graphs (bar, line, pie)	• Use mock appliances or labels to calculate energy use • Graph energy savings before and after upgrades • Introduce the modeling phase of the STEM project	• Energy calculation worksheet • Group graphing activity • Quiz: “Understanding your electricity bill”	• Sample appliance power ratings • Graph paper or digital tools (Excel, Google Sheets) • Mock or real electricity bills

Plan a Solution

Lesson 06: How is heat trapped in the solar water still?

Driving Question:	Lesson Summary:	Learning Outcomes:	Learning Tasks:	Assessment Tasks:	Resources:
How is heat trapped in the solar water still?	Students explore existing solar still technologies and build prototypes in teams. They test them under sunlight to understand how heat is absorbed and retained to produce fresh water.	• Explain how the Sun heats water in a closed solar still using the greenhouse effect. • Identify ways to increase heat absorption in the solar still. • Apply the principles of the greenhouse effect to explain global warming.	• **Team-Based Jigsaw Workstations**: Students rotate through 6 stations, each focusing on a different solar still technology. ➢ Learn from existing models and research how they function. ➢ Discuss design features that maximize heat retention. • **Planning Session**: Students begin planning their own improved solar still prototypes based on what they've learned.	• Observe and assess students' discussion at workstations for scientific understanding • Evaluate teamwork and design thinking using a **Teamwork and Communication Rubric**.	**Reading Handouts for Workstations** • Survival DIY • Solar Water Distiller • Pyramid Solar Still • Spherical Water Still • The Watercone • Aquamate Solar Still • **Prototype Planning Worksheet** • **Materials:** ➢ Large and small empty plastic bottles ➢ Food containers/tubs ➢ Cardboard, cling film, polystyrene and plastic sheets ➢ Aluminum foil, funnels ➢ Pipes (rigid and hose)

					➢ Scotch tape, hot glue guns, scissors, cutters

Try a Solution

Lesson:07 How do we build a prototype of a solar water still?

Driving Question:	Lesson Summary:	Learning Outcomes:	Learning Tasks:	Assessment Tasks:	Resources:
How do we build a prototype of a solar water still?	Teams use their design plans to build solar still prototypes. They apply engineering and teamwork skills to construct functional stills and set them outside to test.	• Execute the planned prototype design using chosen materials. • Provide evidence-based justification for design decisions. • Collaborate effectively as a team during construction	• Finalize design plans (5 mins) • Gather materials and build prototypes (15 mins) • Record design details and calculations (10 mins) • Set up the stills outdoors for sunlight exposure (5 mins)	• Monitor and assess team discussions and construction for scientific reasoning. • Continue using the **Teamwork & Communication Rubric**.	• **Per Team**: Prototype Trial Worksheet, measuring cylinder • **Materials** (same as Lesson 6) • **Outdoor Space** for sunlight exposure

Test a Solution

Lesson 08: How do we test a prototype of a solar water still?

Driving Question	**Lesson Summary**	**Learning Outcomes:**	**Learning Tasks:**	**Assessment Tasks:**	**Resources:**
How do we test a prototype of solar water still?	Students redesign and remake their solar stills and place them in the Sun. They test the efficiency of their improved prototypes. Students also create a poster that explains their design and their plan for safe usage of the resulting brine.	• Execute their plan to build an improved prototype and provide evidence-based reasons to support their design • Collaborate and communicate as a team to build an improved prototype of a solar water still	• In teams, students revise their prototype: ➢ Make the necessary changes ➢ Set up the improved still outdoors ➢ Observe and record data ➢ Calculate the efficiency: Efficiency= Liters of water collected/ m2 of surface area×1 day (10 hours of sunlight)	• Evaluate students' scientific reasoning and teamwork using the **Teamwork and Communication Rubric** • Observe how well students connect design changes to scientific principles • Assess time and task management	• Prototype Trial Worksheet (from Lesson 7) • Measuring cylinder • Outdoor space for sunlight **Materials:** • 5L plastic water bottles, small containers • Cardboard, cling film, polystyrene/plastic sheets • Aluminum foil, funnels, various pipes • Tape, glue guns, scissors, cutters

Communicate solution to the posed problem

Lesson 09: How do we communicate the prototype of solar water still to our client?

Driving Question:	Lesson Summary:	Learning Outcomes	Learning Tasks:	Assessment Tasks:	Resources:
How do we communicate the prototype of solar water still to our client?	Final prototypes and posters are displayed during a **gallery walk**. The **PCRWR** (Pakistan Council of Research in Water Resources) is represented by selected teachers who assess projects. Students' videos are judged for the **Valuing Water Award**.	• Calculate efficiency of their second prototypes • Communicate their design clearly and effectively to the client (PCRWR) • Link design decisions to scientific and mathematical evidence	• Present their prototypes and posters in a common area • Review peer projects and communicate with judges • Teams choose and nominate a winner and a runner-up for the Valuing Water Award	• Evaluate the final **prototype and video** using rubrics • Assess communication clarity, scientific explanation, and evidence use	• Measuring cylinder, chart paper, markers **Facilitated by:** • A group of selected teachers representing PCRWR